Exp: 4

Setup and Configuration of a LAN using a Switch and Ethernet Cables

Aim:

To set up a LAN with 3-4 host machines connected through a switch, assign IP addresses, check connectivity with ping, and share files and folders.

Requirements:

* 3-4 computers
* Switch
* Straight-through Ethernet cables

Procedure:

1. Connect each host machine to a port of the switch using Ethernet cables.
2. Assign IP ~~addresses~~ addresses (Control Panel > Network and Sharing Center > Change adapter settings > Properties > IPv4):
   PC1: 192.168.1.1 PC2: 192.168.1.2 PC3: 192.168.1.3 PC4: 192.168.1.4
   Subnet mask: 255.255.255.0
3. Check connectivity between the machines:
   ping 192.168.1.2
4. Share a folder: right-click the folder > Properties > Sharing > Share, choose the user and permissions (Read/Write).
5. From another machine open Run, type \\192.168.1.1 and access the shared folder and files.

Result:

A LAN was set up using a switch; connectivity was verified with ping and files/folders were shared across the machines.